UNIVERSITY OF BIRMINGHAM

Category Theory

Midlands Graduate School 2026

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Big Picture / Motivation

- Category theory is a “theory”, like set theory, like group theory, like type theory
- Types in functional programming $\rightsquigarrow$ sets (or set-like objects)
- Functions $\rightsquigarrow$ arrows/morphisms between types
- Category theory gives language to talk about how we compose computations, abstract patterns, and universal constructions that mirror programming constructs (pairs, function types, currying)
- It provides means of abstraction to reason and prove things generically, instead of on case-by-case basis

Course Outline

Part I: Sets and Warm-Up

Sets, functions, isomorphisms, quotients and equivalences

Part II: Categories & Functors

Definitions and examples, products, coproducts, commutative diagrams, universal properties

Part III: Natural Transformations

Natural transformations, exponentials, Yoneda Theorem

Part IV: Adjunctions & Free Objects

Adjunctions, free objects, initial algebras, induction

These slides:

www.sergey-goncharov.org/mgs-2026

Recommended Sources

- **Steve Awodey** — *Category Theory*, 2nd ed., Oxford University Press, 2010
  - [accessible introduction]
- **Emily Riehl** — *Category Theory in Context*, Dover, 2016
  - [freely available at https://math.jhu.edu/~eriehl/161/context.pdf]
- **Roy Crole** — *Categories for Types*, Cambridge University Press, 1993
  - [CS-oriented; Category theory for types & denotational semantics]
- **Saunders Mac Lane** — *Categories for the Working Mathematician*, 2nd ed., Springer, 1998
  - [timeless classics]

Sets and Warm-Up

Why Begin with Sets?

- Mathematics builds on sets
- Types in programming follow set intuition
- Category theory (standardly) also builds on sets
- But: what exactly is a set?
    - OK, that is too ambitious – what is it roughly?
- We start with sets to discover something deeper is hiding underneath

Naïve (Philosophical) Sets

- Set is collection of objects, like

$$X = \left\{ \text{⚽}, \text{🐕}, \pi, \mathbb{N}, \heartsuit \right\}$$

- We write $x \in X$ to say that x is member of X, e.g. $\pi \in X$
- Empty set $\varnothing = \{\ \}$ is a set of no elements
- There are various mechanisms to construct larger sets
- Set comprehension

$$\{x \in S \mid P(x)\} \qquad \text{e.g.} \qquad \text{evens} = \{n \in \mathbb{N} \mid \exists m \in \mathbb{N}.\, n = 2m\}$$

Intuition: Sets are containers

Finite Sets v.s. Finite Lists

- (Finite) lists over set X: $\mathrm{List}(X) = \{[x_1, \ldots, x_n] \mid n \in \mathbb{N}, x_1 \in X, \ldots, x_n \in X\}$
- **Notation:** $[\,]$ – empty list, $x :: xs$ – list with head x and tail xs. E.g. $x :: [\,] = [x]$
- Finite sets are lists, where we additionally enable
  - Commutativity: $[\ldots, x, y, \ldots] \equiv [\ldots, y, x, \ldots]$
  - Idempotence: $[\ldots, x, x, \ldots] \equiv [\ldots, x, \ldots]$
- (Finite) Multisets = lists with commutativity, but not idempotence. Thus

Lists < Multisets < Sets

Sets (Slightly More) Formally

- Set theory postulates: everything is a set
- Only primitive relation between sets is '$\in$'

We then derive:

- Natural numbers: $0 = \varnothing$, $1 = \{0, \varnothing\} = \{\varnothing\}$, $2 = \{1, \varnothing\} = \{\{\varnothing\}, \varnothing\}, \ldots$
- Set inclusion: $X \subseteq Y$ if $\forall x \in X.\, x \in Y$
  - Extensionality axiom: $X = Y$ if and only if $X \subseteq Y$ and $Y \subseteq X$
- Powerset: $Y \in \mathcal{P}(X)$ if and only if $Y \subseteq X$
- Relations: $R \subseteq X \times Y$
- Functions: $f : X \to Y$ – such relations $R \subseteq X \times Y$ that for every $x \in X$ there is unique $y \in Y$ with $x \, R \, y$
  - So, in set theory: Sets $\rightsquigarrow$ Relations $\rightsquigarrow$ Functions

Empty Set and Singleton Sets

- No controversy about empty set: $\varnothing = \{\ \}$ – set with no elements
- But what is one-element set? {⚽}? {🐕}? $\{\pi\}$?
- $\{\varnothing\}$ is good candidate
- For every set S, $\{S\}$ is also singleton set
- Singleton sets are characterized by property: they are sets with exactly one element
- $\{\varnothing\}$ is a concrete representative in the class of all singletons

Note on Classes

- All sets do not form a set, but a class

$$\mathsf{Set} < \mathsf{Class}$$

- Division into sets and proper classes is way to avoid paradoxes
- Other approach: universes

$$\mathsf{Set}_0 < \mathsf{Set}_1 < \mathsf{Set}_2 < \dots$$

  where the union of all sets in Set_i is in Set_{i+1}
- By saying "set" we can imagine set from Set_i, and by saying "class" we can imagine set from Set_{i+1}

Natural Numbers as Sets

- In computer science, we count from 0: $\mathbb{N} = \{0, 1, \ldots\}$!
- Von Neumann numerals: $0 := \varnothing, n + 1 := n \cup \{n\}$ So:

$$0 = \varnothing, \quad 1 = \{\varnothing\}, \quad 2 = \{\varnothing, \{\varnothing\}\}, \quad 3 = \{\varnothing, \{\varnothing, \{\varnothing\}\}\}, \ldots$$

  This is very convenient, because $n + 1 = \{0, \ldots, n\}$
- Alternatively Zermelo numerals: $0 := \varnothing, n + 1 := \{n\}$. This is less convenient

Ordered Pairs and Products

We would like to define Cartesian products $X \times Y = \{(x, y) \mid x \in X, y \in Y\}$. But what is (x, y)?

- Kuratowski definition: $(x, y) := \big\{\{x\}, \{x, y\}\big\}$
- Then (PROVE IT) :

$$(x, y) = (x', y') \quad \Longleftrightarrow \quad x = x' \quad \text{and} \quad y = y'.$$

- Using this, we can define:

$$X \times Y := \Big\{\{\{x\}, \{x, y\}\} \subseteq \mathcal{P}(X \cup Y) \mid x \in X,\ y \in Y\Big\}$$

Pairs and Cartesian products are constructed objects!

Disjoint Unions

- Union of sets: $x \in X \cup Y$ iff $x \in X$ or $x \in Y$
- How to define Disjoint Union $X + Y$ (= $X \uplus Y$)?
- Maybe $X + Y = \{(x, \text{🎸}) \mid x \in X\} \cup \{(x, \text{🔨}) \mid y \in Y\}$?

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- More seriously: $X + Y = \{(x, 0) \mid x \in X\} \cup \{(x, 1) \mid y \in Y\}$

Now compute with von Neumann numerals: $1 = \{0\}$, $2 = \{0, 1\}$. Then

$$1 + 1 = \{(1, 0), (1, 1)\} = \big\{\{\{1\}, \{1, 0\}\}, \{\{1\}\}\big\}$$

This is not equal to $2 = \{0, 1\}$! So

$$1 + 1 \neq 2$$

Functions between Sets

- Function $f : X \to Y$, assigns each $x \in X$ a unique $f(x) \in Y$
- Functions $f : X \to Y$ are special relations $f \subseteq X \times Y$
- Identity function $\mathrm{id}_X : X \to X$, $\mathrm{id}_X(x) = x$
- Equality of functions $f = g$ if for all x, $f(x) = g(x)$
  - This is provable principle (PROVE IT) , called function extensionality

Composition:

$$(g \circ f)(x) = g(f(x))$$

Proposition: $f \circ \mathrm{id} = f$, $\mathrm{id} \circ f = f$ and $f \circ (g \circ h) = (f \circ g) \circ h$ (associativity)

Bijection vs Isomorphism

Function $f : X \to Y$ is

- Injection if $f(x) = f(x')$ entails $x = x'$
- Surjection if for every $y \in Y$ there is $x \in X$, such that $f(x) = y$
- Bijection if it is both injection and surjection

Alternative view: $f : X \to Y$ is isomorphism if for some $f^{-1} : Y \to X$,

$$f \circ f^{-1} = \mathrm{id}_Y \quad \text{and} \quad f^{-1} \circ f = \mathrm{id}_X$$

Proposition: f is bijection if and only if f is isomorphism

Some Properties of Isomorphisms

It follows that (PROVE IT)

- f^{-1} is unique if exists
- If f is isomorphism, so is f^{-1}
- id is isomorphism and $\mathrm{id}^{-1} = \mathrm{id}$
- Composition of isomorphisms $f \circ g$ is isomorphism and $(f \circ g)^{-1} = g^{-1} \circ f^{-1}$

These are derivable from definition, without using that isomorphism = bijection!

Notation: $X \cong Y$ if there is isomorphism $f : X \to Y$

Up-to Isomorphism

- Big innovation of category theory: work with things via their descriptions
- We narrow down objects of interest as those that “fit best” to such description
- Such descriptions together with their best-fit condition are called universal properties
- They determine objects not uniquely, but up-to isomorphism

Example. A singleton set 1 is characterised by: there is exactly one function $X \to 1$ for every set X

- $\{\varnothing\}$, $\{\{\varnothing\}\}$, {⚽} all satisfy this — and are all isomorphic
- Any two singletons are isomorphic, so we speak of the singleton up to isomorphism

Induction/Recursion for Natural Numbers

- Set theory postulates existence of natural numbers $\mathbb{N}$ as smallest set that contains $0 = \varnothing$ and closed under $n \mapsto n + 1 := n \cup \{n\}$
- Induction is paraphrasing this: if $P \subseteq \mathbb{N}$ with $0 \in P$ and $n \in P \Rightarrow n{+}1 \in P$, then $P = \mathbb{N}$
  - Because predicates over $\mathbb{N} \iff$ subsets of $\mathbb{N}$:

$$P(0) \wedge (\forall n.\, P(n) \Rightarrow P(n + 1)) \Rightarrow \forall n.\, P(n)$$

- (Structural) Recursion: for any $a \in A$ and $h \colon A \to A$, there is unique $f \colon \mathbb{N} \to A$ with

$$f(0) = a \qquad f(n{+}1) = h(f(n))$$

- Primitive recursion is derivable from recursion (PROVE IT) : for any $g \colon X \to A$ and $h \colon A \times X \times \mathbb{N} \to A$, there is unique $f \colon X \times \mathbb{N} \to A$ with

$$f(x, 0) = g(x) \qquad f(x, n{+}1) = h(f(x, n), x, n)$$

Induction/Recursion for Lists

- Given set X, $\mathrm{List}(X)$ is constructable as smallest set that contains $[\,]$ and closed under $xs \mapsto x :: xs$ for $x \in X$
- Induction: $P([\,]) \wedge (\forall x \in X, xs.\, P(xs) \Rightarrow P(x :: xs)) \Rightarrow \forall xs.\, P(xs)$
- (Structural) Recursion: for any $a \in A$ and $h \colon X \times A \to A$, there is unique $f \colon \mathrm{List}(X) \to A$ with

$$f([\,]) = a \qquad\qquad f(x :: xs) = h(x, f(xs))$$

Questions:

- What is primitive recursion for lists?
- Prove it is derivable
- Define length and concatenation recursively

Remarks:

- Structural recursion $\neq$ general recursion (no arbitrary self-calls)
- Induction — proof principle; recursion — definition principle

Sets and Warm-Up

Equivalences and Quotients

Equivalence Relations

A relation $R \subseteq X \times X$ is an equivalence relation if

- Reflexive: $x \mathrel{R} x$
- Symmetric: $x \mathrel{R} y \Rightarrow y \mathrel{R} x$
- Transitive: $x \mathrel{R} y$ and $y \mathrel{R} z \Rightarrow x \mathrel{R} z$

For $x \in X$, define corresponding equivalence class

$$[x]_R = \{y \in X \mid x \mathrel{R} y\}$$

Equivalence classes partition X: $X = \bigcup_{x \in X} [x]_R$, $[x]_R \cap [y]_R \neq \varnothing$ iff $x \mathrel{R} y$

Quotient Sets

- Given equivalence relation $R \subseteq X \times X$, define quotient set

$$X/R = \{[x]_R \mid x \in X\}$$

- There is natural function

$$\pi : X \to X/R, \qquad \pi(x) = [x]_R$$

- Two elements have same image iff they are related:

$$\pi(x) = \pi(y) \quad \Longleftrightarrow \quad x\,R\,y$$

Example: Integers $\mathbb{Z} = (\mathbb{N} \times \mathbb{N}) / \sim$ where $(m, n) \sim (m', n')$ iff $m + n' = m' + n$ (PROVE IT)

Images and Inverse Images

Given $f : X \to Y$,

- For $A \subseteq X$ define direct image

$$f[A] = \{f(x) \mid x \in A\}$$

- For $B \subseteq Y$ define inverse image

$$f^{-1}[B] = \{x \in X \mid f(x) \in B\}$$

- Let $f^{-1}(x) = f^{-1}[\{x\}]$ – these are often called fibers (over $x \in Y$)

Equivalence Relations from Functions

- Given function $f : X \to Y$, kernel equivalence $\sim_f \subseteq X \times X$ is as follows:

$$x \sim_f y \quad \iff \quad f(x) = f(y)$$

  This is indeed equivalence relation (PROVE IT)

- Equivalence classes are precisely the fibres

$$[x]_{\sim_f} = f^{-1}(f(x))$$

Surjections are Equivalent to Equivalences

- Start with equivalence $R \subseteq X \times X$, then with $\pi : X \to X/_R$

$$R \;\cong \sim_\pi$$

— every equivalence is isomorphic to kernel equivalence of induced quotienting map (PROVE IT)

- Start with surjection $f : X \to Y$, then

$$Y \cong X/\sim_f$$

— codomain of any surjection is isomorphic to quotient of domain by kernel equivalence (PROVE IT)

Example. Let $f : \mathbb{N} \times \mathbb{N} \to \mathbb{Z}$, $f(m, n) = m - n$. Then:

$$(m, n) \sim_f (m', n') \iff m - n = m' - n' \iff m + n' = m' + n$$

— exactly the equivalence used to construct $\mathbb{Z} = (\mathbb{N} \times \mathbb{N})/\sim$

Choice

When we say

"Codomain of any surjection is isomorphic to quotient of domain by kernel equivalence"

there is a snag

To build isomorphism, we need axiom of choice to identify concrete x in equivalence class $[x]_{\sim_f}$

Without that, the equivalence becomes definition: it defines exactly those surjections that are quotient maps of equivalence relations

Category theory calls them regular epimorphisms

Categories & Functors

Definition of Category

Category $\mathcal{C}$ consists of:

- Class of objects $\mathrm{Ob}(\mathcal{C})$
- For every A, B set of morphisms $\mathrm{Hom}_{\mathcal{C}}(A, B)$. Write $f : A \to B$
- Composition: if $f : A \to B$ and $g : B \to C$ then $g \circ f : A \to C$
- Identity morphisms $\mathrm{id}_A : A \to A$

Subject to associativity $(h \circ g) \circ f = h \circ (g \circ f)$ and identity laws $\mathrm{id} \circ f = f = f \circ \mathrm{id}$

We can imagine categories as (multi)graphs: objects = nodes, morphisms = edges

Sets as Category

Category of sets **Set**

- **Objects:** sets
- **Morphisms:** functions between sets
- Composition: usual function composition
- Identities: identity functions

Further set-like categories (objects = sets, identities = identity functions):

- Relations (**Rel**) – morphisms are relations
- Partial functions
- Injections
- Isomorphisms

  ...

Category Pos

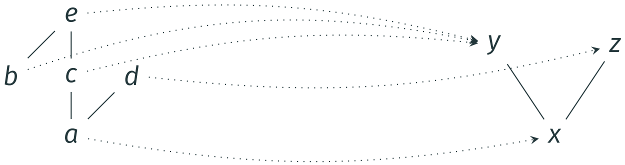
- **Objects:** partially ordered sets $(P, \leq)$
- **Morphisms:** monotone functions

$$f\colon (P, \leq_P) \to (Q, \leq_Q)$$

such that

$$x \leq_P y \Rightarrow f(x) \leq_Q f(y)$$

Example:



Category Mon

- **Objects:** monoids $(M, \cdot, e)$
- **Morphisms:** monoid homomorphisms are maps $f : M \to N$, such that unit and multiplication are preserved:

$$f(x \cdot y) = f(x) \cdot f(y) \qquad f(e_M) = e_N$$

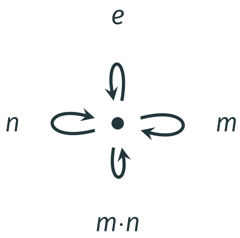
Example: $\mathrm{length} : (\mathrm{List}(A), +\!\!+, [\,]) \to (\mathbb{N}, +, 0)$ — preserves unit and multiplication:

$$\begin{aligned} \mathrm{length}([\,]) &= 0 \\ \mathrm{length}(xs +\!\!+ ys) &= \mathrm{length}(xs) + \mathrm{length}(ys) \end{aligned}$$

Monoid as One-Object Category

Fix some monoid $(M, \cdot, e)$. We can turn it into category!

- **Objects:** — exactly one, say $\bullet$
- **Morphisms:** $\mathrm{Hom}(\bullet, \bullet) = M$
- Composition is given by monoid multiplication
- Identity given by e



This is equivalence: Monoids $\iff$ one-object categories (PROVE IT)

Monoid homomorphisms $\iff$ functors between one-object categories (PROVE IT)

Posets as Categories

Fix poset $(P, \leq)$. It is also category:

- **Objects:** elements of P
- **Morphisms:** $\mathrm{Hom}(x, y) = \{\star\}$ if $x \leq y$ and $\mathrm{Hom}(x, y) = \varnothing$ otherwise (at most one morphism between two objects)
- Unit of composition = reflexivity
- Associativity of composition = transitivity

Hint: Categorical notions become order-theoretic when instantiated to poset-categories.
To understand a complex categorical notion, instantiate it to poset-category!

Example Categories: Summary

Category	Objects	Morphisms	Composition
Set	sets	functions	function composition
Pos	posets	monotone maps	function composition
Mon	monoids	homomorphisms	function composition
Rel	sets	relations $R \subseteq X \times Y$	relational composition
$(P, \leq)$	elements of P	$\star$ if $x \leq y$	transitivity
$(M, \cdot, e)$	$\{\star\}$	elements of M	monoid multiplication

Commutative Diagrams

- For $f : A \to B$, we can express $f \circ \mathrm{id}_A = f$ and $\mathrm{id}_B \circ f = f$ diagrammatically:

$$\begin{tikzcd} & A \arrow[dl, equal] \arrow[d, "f"] \arrow[dr, equal] & \\ A \arrow[r, "f"] & B & A \arrow[l, "f"'] \end{tikzcd}$$

- Associativity:

$$\begin{tikzcd} A \arrow[r, "f"] \arrow[rr, bend left, "g \circ f"] & B \arrow[r, "g"] \arrow[rr, bend right, "h \circ g"'] & C \arrow[r, "h"] & D \end{tikzcd}$$

But that is redundant, because we read sequence f, g as $g \circ f$ – associativity is baked in!

Isomorphisms

Morphism $f : A \to B$ is an isomorphism if there exists $f^{-1} : B \to A$ with $f^{-1} \circ f = \mathrm{id}_A$ and $f \circ f^{-1} = \mathrm{id}_B$:

$$\begin{tikzcd} A \arrow[loop left, "\mathrm{id}_A"] \arrow[r, bend left, "f"] & B \arrow[l, bend left, "f^{-1}"] \arrow[loop right, "\mathrm{id}_B"] \end{tikzcd}$$

- A and B are isomorphic, written $A \cong B$, if such f exists
- f^{-1} is unique: if $g \circ f = \mathrm{id}_A$ and $f \circ g = \mathrm{id}_B$, then

$$g = g \circ \mathrm{id}_B = g \circ (f \circ f^{-1}) = (g \circ f) \circ f^{-1} = \mathrm{id}_A \circ f^{-1} = f^{-1}$$

- In **Set**: isomorphism = bijection; in **Pos**: order-isomorphism; in **Mon**: monoid isomorphism
- Group is one-object category in which every morphism is isomorphism (PROVE IT)

Commutative Diagrams and Equational Reasoning

A diagram commutes if every directed path with the same start and end gives the same composite.

$$\begin{tikzcd}
A \arrow[r, "f"] \arrow[d, "a"'] & B \arrow[r, "g"] \arrow[d, "b"] & C \arrow[d, "c"] \\
D \arrow[r, "h"'] & E \arrow[r, "k"'] \arrow[d, "p"'] & F \arrow[d, "q"] \\
& G \arrow[r, "r"'] & H
\end{tikzcd}$$

If all three squares commute:

$$b \circ f = h \circ a, \quad c \circ g = k \circ b, \quad q \circ k = r \circ p$$

then the outer paths from A to H agree:

$$\begin{aligned}
q \circ c \circ g \circ f &= q \circ (k \circ b) \circ f \\
&= (q \circ k) \circ (b \circ f) \\
&= (r \circ p) \circ (h \circ a) \\
&= r \circ p \circ h \circ a
\end{aligned}$$

Thus, commutative diagrams = pictorial representations of equational proofs!

Not Every Diagram Proves Something

$$\begin{tikzcd}
A \arrow[rr, "f"] \arrow[dd, "g"'] & & B \arrow[d, "h"] \\
& C \arrow[ul, "u"'] \arrow[r, "k"] \arrow[d, "j"'] & D \arrow[d, "r"] \\
E \arrow[r, "p"] & F \arrow[r, "q"] & G
\end{tikzcd}$$

Commuting cells (by assumption):

$$h \circ f \circ u = k$$

$$p \circ g \circ u = j$$

$$r \circ k = q \circ j$$

Chaining all three:

$$\begin{aligned}
(r \circ h \circ f) \circ u &= r \circ k \\
&= q \circ j \\
&= (q \circ p \circ g) \circ u
\end{aligned}$$

But this does not imply $r \circ h \circ f = q \circ p \circ g$, because we generally cannot discard u!

Terminal and Initial Objects

In category $\mathcal{C}$:

- Terminal object 1: for every object A there is exactly one morphism $A \to 1$
- Initial object 0: for every object A there is exactly one morphism $0 \to A$

Examples:

- In **Set**: any singleton $\{*\}$ is terminal; $\varnothing$ is initial
- In **Pos**: any one-element poset is terminal; $\varnothing$ is initial
- In **Mon**: trivial monoid $\{e\}$ is terminal and initial
- In every poset category: terminal = greatest element; initial = least element

Key viewpoint: elements of X in **Set** correspond to arrows $1 \to X$, i.e. elements$(X) \cong \mathrm{Hom}(1, X)$. So, $f(x)$ must be interpreted as composition $1 \xrightarrow{x} X \xrightarrow{f} Y$!

Product Universal Property

- **Object:** a product of X and Y is an object Z
- **Diagram (span):** morphisms $p_1 : Z \to X$ and $p_2 : Z \to Y$
- **Universal property:** for any $f : A \to X$ and $g : A \to Y$ there is a unique $h : A \to Z$ with $p_1 \circ h = f$ and $p_2 \circ h = g$:

$$
\begin{tikzcd}
 & A \arrow[dl, "f"'] \arrow[d, dashed, "h"] \arrow[dr, "g"] & \\
X & Z \arrow[l, "p_1"] \arrow[r, "p_2"'] & Y
\end{tikzcd}
$$

Selected product $X \times Y$: specific choice of product for every pair (X, Y); all products of X and Y are uniquely isomorphic

Notation: With $X \times Y$, use $\mathrm{fst}, \mathrm{snd}$ for the chosen projections; $\langle f, g \rangle$ for the induced h

Examples of Products

- **Set:** $X \times Y = \{(x, y) \mid x \in X,\ y \in Y\}$; projections $(x, y) \mapsto x$ and $(x, y) \mapsto y$; pairing $\langle f, g\rangle(x) = (f(x), g(x))$
- **Pos:** $X \times Y$ with pointwise order: $(x, y) \leq (x', y')$ iff $x \leq x'$ and $y \leq y'$
- **Mon:** componentwise monoid $(M \times N,\ \cdot,\ (e_M, e_N))$ where

$$(m, n) \cdot (m', n') = (m \cdot m',\ n \cdot n')$$

- **Poset category:** product of x and y = meet $x \wedge y$ (=greatest lower bound), when it exists

Products are Up-To-Isomorphism

Let (Z, p_1, p_2) and (Z', p_1', p_2') both be products of X and Y:

- Universal property of Z: $\exists!\, i\colon Z' \to Z$ with $p_1 \circ i = p_1'$, $p_2 \circ i = p_2'$
- Universal property of Z' on span (p_1, p_2): $\exists!\, j\colon Z \to Z'$ with $p_1' \circ j = p_1$, $p_2' \circ j = p_2$
- By uniqueness: $j \circ i = \mathrm{id}_{Z'}$
- By symmetric argument: $i \circ j = \mathrm{id}_Z$, so $Z \cong Z'$

$$\begin{tikzcd}
& Z' \arrow[dl, "p_1'"'] \arrow[dr, "p_2'"] \arrow[d, dashed, "i"] & \\
X & Z \arrow[l, "p_1"'] \arrow[r, "p_2"] \arrow[d, dashed, "j"] & Y \\
& Z' \arrow[ul, "p_1'"] \arrow[ur, "p_2'"'] &
\end{tikzcd}$$

Example: Take selected product $(X \times Y, \mathrm{fst}, \mathrm{snd})$. Then $(Y \times X, \mathrm{snd}, \mathrm{fst})$ is also product of X and Y – isomorphism swap $= \langle \mathrm{snd}, \mathrm{fst} \rangle : X \times Y \xrightarrow{\sim} Y \times X$

Axioms of Binary Products

For selected products, one can show (PROVE IT) :

$$(\beta_1) \quad \mathsf{fst} \circ \langle f, g \rangle = f$$

$$(\beta_2) \quad \mathsf{snd} \circ \langle f, g \rangle = g$$

$$(\eta) \quad \langle \mathsf{fst}, \mathsf{snd} \rangle = \mathsf{id}_{X \times Y}$$

$$(\mathrm{nat}) \quad \langle f, g \rangle \circ h = \langle f \circ h, \ g \circ h \rangle$$

Theorem: This is a sound and complete axiomatization of binary products (PROVE IT)

Example (uniqueness of pairing): if $\mathsf{fst} \circ k = f$ and $\mathsf{snd} \circ k = g$, then

$$k \ = \ \mathsf{id} \circ k \ \stackrel{(\eta)}{=} \ \langle \mathsf{fst}, \mathsf{snd} \rangle \circ k \ \stackrel{(\mathrm{nat})}{=} \ \langle \mathsf{fst} \circ k, \ \mathsf{snd} \circ k \rangle \ \stackrel{(\mathrm{assm.})}{=} \ \langle f, g \rangle$$

Dual Category

Given category $\mathcal{C}$, the opposite category $\mathcal{C}^{\text{op}}$ has:

- Same objects as $\mathcal{C}$
- Morphisms reversed: $f : A \to B$ in $\mathcal{C}$ becomes $f^{\text{op}} : B \to A$ in $\mathcal{C}^{\text{op}}$
- Composition: $g^{\text{op}} \circ f^{\text{op}} = (f \circ g)^{\text{op}}$

Principle of Duality: any theorem proved in $\mathcal{C}$ yields a dual theorem in $\mathcal{C}^{\text{op}}$ — just reverse all arrows

Examples:

- Poset $(P, \leq)^{\text{op}}$ = poset with reversed order $(P, \geq)$
- Initial objects in $\mathcal{C}$ = terminal objects in $\mathcal{C}^{\text{op}}$

Coproduct via Duality

Coproduct = product in $\mathcal{C}^{\mathrm{op}}$ — obtained by reversing all arrows in the product definition

- **Object:** a coproduct of X and Y is an object Z
- **Diagram (cospan):** morphisms $\iota_1 : X \to Z$ and $\iota_2 : Y \to Z$
- **Universal property:** for any $f : X \to A$ and $g : Y \to A$ there is a unique $h : Z \to A$ with $h \circ \iota_1 = f$ and $h \circ \iota_2 = g$:

$$\begin{tikzcd} & A & \\ X \arrow[ur, "f"] \arrow[r, "\iota_1"'] & Z \arrow[u, dashed, "h"] & Y \arrow[ul, "g"'] \arrow[l, "\iota_2"] \end{tikzcd}$$

Selected coproduct $X + Y$**:** choice of coproduct for every pair; notation inl, inr for chosen injections; $[f, g]$ — co-pairing — for the induced h

Example: In **Set**, $X + Y$ = disjoint union (PROVE IT)

Axioms of Binary Coproducts

For selected coproducts, one can show:

$$(\beta_1)\quad [f, g] \circ \mathsf{inl} = f$$

$$(\beta_2)\quad [f, g] \circ \mathsf{inr} = g$$

$$(\eta)\quad [\mathsf{inl}, \mathsf{inr}] = \mathsf{id}_{X+Y}$$

$$(\mathrm{nat})\quad h \circ [f, g] = [h \circ f,\ h \circ g]$$

Theorem: This is sound and complete axiomatization of binary coproducts

Proof: Duality

Example: Every $f : X + Y \to Z$ has form $[f_1, f_2]$ for $f_1 : X \to Z$, $f_2 : Y \to Z$. Indeed:

$$f \ = \ f \circ \mathsf{id} \overset{(\eta)}{=} f \circ [\mathsf{inl}, \mathsf{inr}] \overset{(\mathrm{nat})}{=} [f \circ \mathsf{inl},\ f \circ \mathsf{inr}]$$

Categories & Functors

Functors

Definition of Functor

- (Covariant) Functor $F \colon \mathcal{C} \to \mathcal{D}$ assigns:
  - To every object $A \in \mathrm{Ob}(\mathcal{C})$ an object $F(A) \in \mathrm{Ob}(\mathcal{D})$
  - To every morphism $f : A \to B$ a morphism $F(f) \colon F(A) \to F(B)$

  preserving composition and identities:

$$F(g \circ f) = F(g) \circ F(f) \qquad F(\mathrm{id}_A) = \mathrm{id}_{F(A)}$$

- Functor $F \colon \mathcal{C} \to \mathcal{C}$ is called endofunctor
- Functors compose: $(GF)(A) = G(F(A))$, $(GF)(f) = G(F(f))$; categories and functors form the category **Cat** — category of categories modulo size issues (sets vs. classes)

Examples of Functors

Endofunctors on Set:

- Powerset: $\mathcal{P}(X)$; on $f\colon X \to Y$ acts as direct image $\mathcal{P}(f) = f[-]$
- List: List(X) = finite lists over X; $\mathrm{List}(f)([x_1, \ldots, x_n]) = [f(x_1), \ldots, f(x_n)]$
- Maybe: $X \mapsto X + \{\mathrm{Nothing}\}$; $\mathrm{Maybe}(f)(\mathrm{inl}\, x) = \mathrm{inl}\, f(x)$, $\mathrm{Maybe}(f)(\mathrm{inr}\, x) = \mathrm{inr}\, x$

Between different categories:

- Forgetful functor $U\colon \mathbf{Mon} \to \mathbf{Set}$: sends $(M, \cdot, e)$ to M, homomorphisms to functions
- Free monoid functor $F\colon \mathbf{Set} \to \mathbf{Mon}$: sends X to List(X)

Product Category and Bifunctors

Product category $\mathcal{C} \times \mathcal{D}$**:** objects are pairs (A, B); morphisms $(A, B) \to (A', B')$ are pairs (f, g) of morphisms $f : A \to A'$, $g : B \to B'$; composition componentwise

Bifunctor $F : \mathcal{C} \times \mathcal{D} \to \mathcal{E}$: functor on product category. Unraveling definition:

- $F(\mathrm{id}_A, \mathrm{id}_B) = \mathrm{id}_{F(A,B)}$
- $F(f' \circ f,\ g' \circ g) = F(f', g') \circ F(f, g)$

Example: binary product $(- \times -) : \mathcal{C} \times \mathcal{C} \to \mathcal{C}$ (if $\mathcal{C}$ has binary products): on morphisms $f : A \to A'$, $g : B \to B'$, gives

$$f \times g = \langle f \circ \mathrm{fst},\ g \circ \mathrm{snd} \rangle : A \times B \to A' \times B'$$

Contravariant and Mixed-Variance Functors

Contravariant functor $F \colon \mathcal{C}^{\mathrm{op}} \to \mathcal{D}$: sends $f : A \to B$ to $F(f) \colon F(B) \to F(A)$; reverses composition: $F(g \circ f) = F(f) \circ F(g)$

Mixed variance $F \colon \mathcal{C}^{\mathrm{op}} \times \mathcal{D} \to \mathcal{E}$: contravariant in first argument, covariant in second

Example: $\mathrm{Hom}(-,-) \colon \mathcal{C}^{\mathrm{op}} \times \mathcal{C} \to$ **Set** is mixed-variance (see next slide)

Duality: contravariant functor $\mathcal{C}^{\mathrm{op}} \to \mathcal{D}$ is the same as covariant functor $\mathcal{C} \to \mathcal{D}^{\mathrm{op}}$; choice of perspective is a matter of convention

Functors: Further Examples

Hom-functor $\mathrm{Hom}(-,-)\colon \mathcal{C}^{\mathrm{op}} \times \mathcal{C} \to \mathbf{Set}$:

- $\mathrm{Hom}(A,-)\colon \mathcal{C} \to \mathbf{Set}$ — covariant; acts by postcomposition: $h \mapsto f \circ h$
- $\mathrm{Hom}(-,B)\colon \mathcal{C}^{\mathrm{op}} \to \mathbf{Set}$ — contravariant; acts by precomposition: $h \mapsto h \circ f$
- For $f : A' \to A$ and $g : B \to B'$, $\mathrm{Hom}(f,g)$ sandwiches input between f and g:

$$\mathrm{Hom}(f,g)\colon \mathrm{Hom}(A,B) \to \mathrm{Hom}(A',B'), \qquad h \mapsto g \circ h \circ f$$

Contravariant powerset sends $X \mapsto \mathcal{P}(X)$ and $f : X \to Y$ to $f^{-1}[-]\colon \mathcal{P}(Y) \to \mathcal{P}(X)$

This is the same as $\mathrm{Hom}_{\mathbf{Set}}(X, 2)$ (**PROVE IT**)

Full and Faithful Functors

Functor $F\colon \mathcal{C} \to \mathcal{D}$ induces maps on hom-sets:

$$F_{A,B}\colon \mathrm{Hom}_{\mathcal{C}}(A, B) \to \mathrm{Hom}_{\mathcal{D}}(FA, FB), \qquad f \mapsto F(f)$$

- F is faithful if every $F_{A,B}$ is injective
- F is full if every $F_{A,B}$ is surjective
- F is fully faithful if every $F_{A,B}$ is a bijection

Examples:

- Forgetful $U\colon \mathbf{Mon} \to \mathbf{Set}$: faithful (distinct hom'ms remain distinct) but not full (not every function is a homomorphism)
- Inclusion $(P, \leq) \hookrightarrow \mathbf{Set}$ via poset-as-category: faithful (Hom has at most one element), not full in general

To Be Continued …